

PREPARATION OF THERMOCHEMICAL SIMULATION v0 FOR THE NEW PHENOLIC LINER

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Purpose of this document. This report records the reproducible preparation of the first model version for the new motor. It strictly separates the requested geometry from assumptions inherited from the earlier simulations. The native Windows model, mesh, restart chain, and MAPDL preflight are ready, but no transient calculation has been started. The primary goal is to compute, without ambiguous terminology, removed phenolic thickness, remaining thickness, pyrolysis-gas mass, and consumed-carbon mass at every time.

1 Definition of the new motor

1.1 Imposed dimensions

The brief supplied for this study contains three diameters: $D_{\text{Al,out}} = 6.000$ in, $D_{\text{Al,in}} = D_{\text{ph,out}} = 5.625$ in, and $D_{\text{ph,in}} = 5.250$ in. Unlike the first two historical simulations, these dimensions are correctly treated as *diameters* and divided by two before they become radial coordinates.

Table 1. Nominal radial interfaces.

Interface	Diameter [in]	Diameter [mm]	Radius [mm]
Phenolic bore	5.250	133.350	66.6750
Phenolic/aluminium	5.625	142.875	71.4375
Aluminium exterior	6.000	152.400	76.2000

The result is one 4.7625 mm phenolic layer and an aluminium shell of the same thickness. There is no epoxy, second phenolic layer, or artificial intermediate contact. The simulation uses a 50 mm axial coupon and a 0.10° sector, giving a 3,600 scale factor to the full ring of the same length. Because the actual motor length was not supplied, no total motor mass is extrapolated beyond this 50 mm segment.

New nominal geometry — radial cross-section

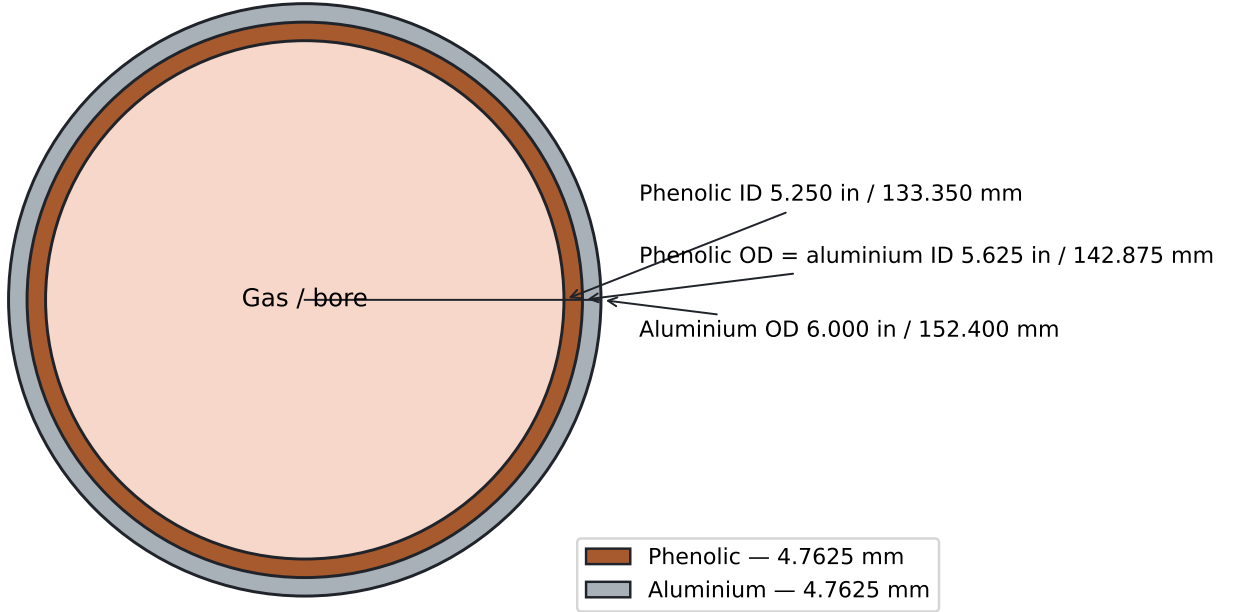


Figure 1. Nominal two-layer cross-section.

As an analytical check, the full 50 mm segment contains $V_{\text{ph},0} = 1.033\,208 \times 10^{-4} \text{ m}^3$, or $m_{\text{ph},0} = 0.129\,151 \text{ kg}$ at the reference density $\rho_v = 1250 \text{ kg m}^{-3}$. The corresponding sector mass is 35.8753 mg. These are balance references, not measurements of the manufactured part.

1.2 Geometry files

The `geometry/newmotor_phenolic_v0.db` database is a native MAPDL 2026 R1 file containing both glued volumes. It was created on Windows without a mesh or a `SOLVE` command. Its standalone, readable generator, `model/generate_geometry_mapdl.inp`, remains the geometry source. The exchange database uses global Z as its axis; the analytical solver mesh uses global Y. A rigid rotation relates them and changes no dimension or thermal condition.

2 Computational domain and mesh

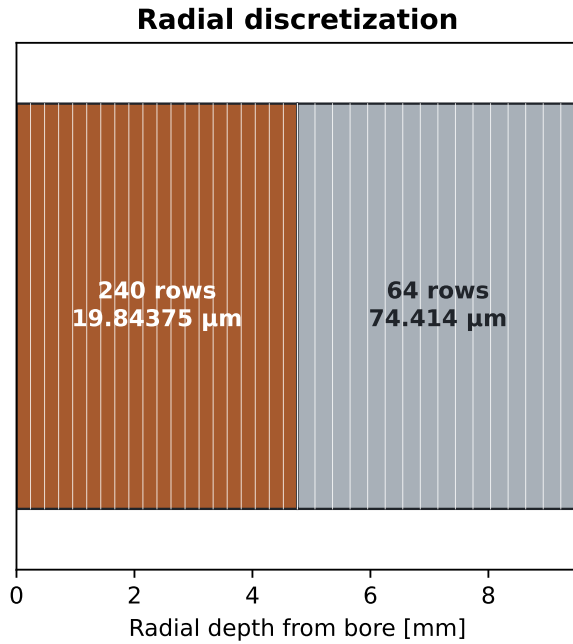
2.1 Discretization choices

The mesh is structured, conformal at the interface, and contains only linear SOLID70 thermal hexahedra. The two materials share interface nodes, so no contact elements are required. The phenolic has 240 radial rows:

$$\Delta r_{\text{ph}} = \frac{4.7625 \text{ mm}}{240} = 19.84375 \mu\text{m}.$$

The aluminium has 64 rows of $74.4141 \mu\text{m}$. The axis is divided into 250 intervals of 0.20 mm and the angle into two intervals. Each potentially exposed phenolic row therefore has 500 faces.

Conformal structured mesh validated in MAPDL



Mesh inventory

Parameter	Value
Angle	0.10°
Axial	250 × 0.20 mm
Circumferential	2 divisions
Nodes	229 665
Hexahedra	152 000
Faces per row	500
Full-ring scale	× 3 600

Figure 2. Radial resolution and final mesh inventory.

The final model has 229,665 nodes and 152,000 hexahedra: 120,000 in phenolic and 32,000 in aluminium. The first 100-division axial mesh produced an aspect ratio of 25.2 and was archived in the temporary directory. Increasing to 250 divisions reduces the ratio to approximately 10. The final native MAPDL preflight completes with zero warnings and zero errors.

2.2 Why a 0.10-degree sector

Two angular divisions retain three-dimensional topology and explicit convection faces, while 240 radial rows provide the resolution needed for progressive removal. The sector cannot represent large-scale circumferential nonuniformity. It is appropriate for a local axisymmetric response; any crack, joint, or localized jet requires a wider angular domain.

3 Planned thermochemical model

3.1 Phenolic pyrolysis

The UserMatTh law retains conversion state variables and returns, for every live element, conversion α , gas production, and energy terms. Virgin and char reference densities are 1250 kg m^{-3} and 600 kg m^{-3} ; the provisional pyrolysis enthalpy is 418 kJ kg^{-1} . Thermal properties depend on temperature and conversion.

The hot face defaults to $T_g = 2230.85^\circ\text{C}$ and $h_0 = 1000 \text{ W m}^{-2} \text{ K}^{-1}$. Blowing continuously reduces the film coefficient as gas flux increases. The outer face exchanges with a 22°C ambient by convection ($h = 8 \text{ W m}^{-2} \text{ K}^{-1}$) and radiation ($\varepsilon = 0.25$).

Qualification limit. Chamber pressure, gas composition, and carbon oxidation kinetics are not yet available. The char-consumption law is therefore explicitly marked **EXPLORATORY_NOT_CALIBRATED**. Results may compare variants and verify balances, but become predictive only after calibration to the actual material and motor.

3.2 Row removal

A row is deactivated only after its residual solid mass, following pyrolysis and carbon consumption, falls below 0.1 % of its initial virgin mass. Removal occurs after acceptance of a converged step and is reapplied after checkpoint restoration. This replaces the fragile decision based only on average conversion.

The nominal coupling step is 0.05 s; the solver may reduce it to 0.0005 s. The 7 s reference horizon is inherited solely for comparison with the earlier study. A new version must replace it before launch if the new motor has a different actual burn duration.

4 Primary quantities and balances

4.1 Remaining thickness

After $N_k(t)$ rows have been accepted for removal,

$$s_{\text{removed}}(t) = N_k(t) \Delta r_{\text{ph}}, \quad s_{\text{remaining}}(t) = s_0 - s_{\text{removed}}(t), \quad s_0 = 4.7625 \text{ mm}.$$

The discrete thickness resolution is therefore 19.843 75 μm . The hot-wall coordinate is $R_h(t) = R_i + s_{\text{removed}}(t)$.

4.2 Consumed phenolic mass

Three quantities must not be conflated:

1. cumulative gas mass produced by pyrolysis;
2. carbon mass consumed at the exposed surface;
3. virgin-equivalent mass of geometrically removed rows.

For a full ring of length L , the last quantity is

$$m_{\text{removed},0}(t) = \rho_v \pi L \left[(R_i + s_{\text{removed}}(t))^2 - R_i^2 \right].$$

The history retains both sector values and values multiplied by 3,600. Every step closes mass between initial mass, remaining solid, pyrolysis gas, consumed carbon, and ejected residual. An error beyond tolerance automatically prevents the next solve.

4.3 Planned outputs

The `newmotor_v0_history.csv` file is atomically rebuilt after every accepted step. It includes time, row index, removed and remaining thickness, sector and full-ring masses, surface temperature, α_{mean} , α_{min} , film coefficient, powers, balance residuals, and warnings. One immutable JSON record is retained for each checkpoint; the CSV is only a derived view of those records.

Result plots will be generated from these converged states: remaining and removed thickness versus time, consumed-mass components, recession rate, temperature and blowing, radial temperature/conversion profiles, and distance from the surface to the 90, 95, and 99 % pyrolysis fronts. A vertical line will mark the first numerically meaningful recession front.

5 Windows execution, pause, and resume

All large files are isolated in Git-ignored `v0/ansystmp`. The `v0` directory is mapped to drive `V:` in Windows, and the runtime is `V:\textbackslash ansystmp\textbackslash windows`. It has no reference to the older project. The UserMatTh source and DLL are copied locally.

Before the first calculation, `launch_v0.ps1 -Preflight` verifies input hashes, mesh, the local `1055@localhost` license, and reads the complete deck without `SOLVE`. The September 20, 2026 preflight confirms 229,665 nodes, 152,000 elements, zero warnings, and zero errors. No solver process is currently active, and no simulation `state.json` has been created.

Launch requires explicit `-Run` and uses 10 DMP ranks. The script refuses to start when ANSYS/MPI already exists. Six restart generations are retained. `request_pause.ps1` requests a stop at the next accepted macro-step; `-Run -Resume` is allowed only from a `PAUSED_AT_CHECKPOINT` state whose inputs remain unchanged.

6 Preparation status and next decisions

Item	Status
Two-volume geometry	created and archived in a native MAPDL database
Structured mesh	validated: 229,665 nodes, 152,000 hexahedra
Windows preflight	complete, 0 warnings, 0 errors, no solve
Solver inputs	sealed with SHA-256
Pause/resume	prepared at accepted checkpoints
French/English reports	compiled from the same parameters
Transient calculation	not started; explicit authorization still required

Before launch, the actual burn duration, chamber pressure, gas composition, and measured phenolic properties remain desirable. Without them, v0 is deliberately a seven-second exploratory reference. Its structure is nonetheless ready to answer the primary question: how much phenolic was transformed or removed, and how much sound thickness remains at each time?